



Original Article

EAST plasma equilibrium reconstruction based on weight-adaptive physics-informed neural network

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ABSTRACT

This paper introduces a weight-adaptive physics-informed neural network (PINN) for plasma equilibrium reconstruction in the EAST tokamak, overcoming the reliance of conventional approaches on predefined current profile assumptions. The proposed framework incorporates the Grad-Shafranov (G-S) equation as a physical constraint and integrates external magnetic diagnostics – such as flux loops and magnetic probes – to jointly exploit both measured data and underlying physics. Three dedicated subnetworks are designed to predict the plasma flux, current profile functions (p' and ff'), and poloidal field coil currents, substantially improving reconstruction accuracy over earlier PINN-based attempts. A hybrid Adam-LBFGS optimizer combined with an adaptive loss weighting scheme ensures robust convergence and higher precision relative to standard PINN. When applied to experimental EAST data, the method demonstrates its capability to accurately match diagnostic measurements while maintaining a low physics loss residual, underscoring the effectiveness of the weight-adaptive strategy. Furthermore, accurate equilibrium reconstruction throughout three full discharges with different shapes confirms the robustness and versatility of the method across diverse plasma configurations. These results demonstrate that the proposed PINN framework can provide a reliable and effective alternative for analyzing EAST discharges.

1. Introduction

Plasma equilibrium reconstruction yields a set of fundamental physical parameters, such as plasma shape, boundary, toroidal current density, beta, and safety factor. These parameters are vital for tokamak operation and data analysis, especially when direct measurement is nearly unattainable. A variety of numerical computation codes have been developed and applied to reconstruct the plasma equilibrium on the tokamaks worldwide, including Equinox on JET and ToreSupra [1], CEDRES++ on ITER, JT-60SA, Tore Supra and WEST [2], NICE on ITER [3], CLISTE [4] and GPEC [5] on ASDEX Upgrade, LIUQE on TCV [6], EFIT++ on MAST [7], PEFIT on EAST, DIII-D and ITER [8], and the highly favored EFIT on DIII-D [9–11], JET [12], NSTX [13], EAST [14], KSTAR [15], and HL-2 A [16]. The computational principles underlying these codes are fundamentally similar. They seek to obtain a plasma equilibrium that satisfies the Grad-Shafranov equation [1–16] by finding the least square best fit to a set of experimental measurements, such as external magnetic measurements and internal

motional stark effect (MSE) diagnostics, under the constraints of a given set of poloidal field coils and device structures. This is an inverse problem based on the Grad-Shafranov (G-S) equation that necessitates solving the nonlinear elliptic partial differential equation (PDE). The equilibrium solutions for the poloidal flux and the toroidal current density of the plasma are derived using a grid covering the entire area of the vacuum vessel. Typically, the plasma current is modeled using a parameterized function (e.g., a low-order polynomial or a tension spline) whose parameters are defined at specified grid points. These modeling assumptions present significant challenges to the reconstruction of high-performance plasma discharges with complex profiles. Furthermore, numerical codes require an iterative computation due to the nonlinearity of the G-S equation, as it is an inverse free-boundary problem in plasma equilibrium reconstruction.

With the advancement of artificial intelligence technology, particularly in deep learning, numerous strategies have been proposed for neural networks to learn and reconstruct the plasma equilibrium in

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tokamaks. On NSTX-U, two neural networks have been developed that can utilize magnetic diagnostics as inputs for equilibrium prediction, serving as an efficient reconstruction code [17]. This method offers some performance enhancements compared to the real-time reconstruction code rEFIT [18]. On KSTAR, a deep neural network has been developed to serve as a G-S solver. Trained on the database generated by offline EFIT and constrained by measured magnetic signals, this network can reconstruct magnetic equilibria in approximately 1 ms [19]. It achieves high accuracy comparable to that of offline EFIT and can handle missing input data through an imputation scheme, demonstrating its potential for real-time plasma control. For the HL-3 tokamak, a multi-task learning neural network that uses magnetic diagnostics has been developed for real-time equilibrium reconstruction [20]. This network enables the simultaneous output of key parameters and high-resolution current density and flux distributions, while enhancing accuracy, computational efficiency, and robustness in generalizing to unseen data.

More significantly, the Physics-Informed Neural Network (PINN) integrates partial differential equations (PDEs) into deep learning through physics-constrained loss functions. This approach enables simultaneous data assimilation and solution of forward/inverse problems while ensuring physical consistency and enhanced generalization, even with sparse datasets [21,22]. Recent scientific computing has witnessed the wide adoption of PINN across diverse domains including fluid mechanics [23], heat transfer [24], power systems [25], nuclear reactor transients [26], metal additive manufacturing [27], multiaxial fatigue life prediction [28], and earthquake prediction [29]. Inspired by the success of PINN in handling complex nonlinear PDEs, studies have emerged to solve the G-S equation for tokamak plasma equilibrium using a PINN framework [30,31]. In a typical PINN implementation, the neural network is optimized by minimizing the residuals of the G-S equation. Ref. [30,31] demonstrate the potential of PINN method in plasma equilibrium reconstruction with external magnetic measurements in KSTAR and JET respectively. Ref. [32] investigates the PINN method to solve the fixed-boundary plasma equilibrium with fully data-free and mesh-free surrogate models of G-S equation. Ref. [33] illustrates the significant computational efficiency of PINN method in solving the free-boundary plasma equilibrium problem.

This work aims to establish a plasma equilibrium reconstruction algorithm for EAST using PINN. Similar to Refs. [30,31], the G-S equation is employed as the physical constraint, while external magnetic measurements, such as flux loops and magnetic probes, are utilized to define the boundary conditions. The proposed PINN framework features a dedicated architecture comprising three subnetworks to predict the poloidal flux, the plasma current profile functions (p' and $f f'$), and the poloidal field (PF) coil currents, respectively (details in Sections 2.2 and 2.3). A key improvement within this architecture lies in its enhanced representation of the PF coil currents. Instead of treating them as fixed known values [30] or assuming a predefined PF current density distribution [31], a dedicated neural network is integrated to dynamically refine the PF currents within the PINN optimization. Consequently, this integrated architecture is designed to reconstruct EAST plasma equilibria that more accurately reflect real discharge experiments. Additionally, achieving robust and physically consistent reconstructions using only basic boundary conditions is challenging due to the inherent ill-posed nature of the inverse problem and measurement noise. To provide stronger coupling between p' , $f f'$ and the magnetic diagnostics, we incorporate Green's function-based response matrices as an enhanced set of physical constraints within the PINN loss function. These constraints, detailed in Section 2.3, effectively correlate the internal current source with the diagnostic measurements, guiding the network towards solutions that are both data-consistent and physically plausible.

In addition to this architectural innovation, effective model training is another critical factor for success. The convergence behavior and accuracy of the vanilla PINN [21,22] algorithm are highly susceptible

to the manually set weights assigned to its constituent loss terms. Recent research confirms that a strategic counterbalance of these terms with appropriate weighting can effectively accelerate model convergence and enhance precision [34]. To this end, this work integrates an adaptive loss weighting technique into the PINN framework for plasma equilibrium reconstruction. Furthermore, the learning rates of these adaptive weights are dynamically adjusted in response to the convergence trend of the physics-informed loss. This implementation yields demonstrably faster convergence and higher accuracy compared to the vanilla PINN approach.

This paper is organized as follows. Section 2 introduces the architecture of weight-adaptive PINN for EAST plasma equilibrium reconstruction, detailing the three specialized subnetworks, loss function construction, and the weight-adaptive strategy. Section 3 presents the comprehensive validation of the proposed method, including benchmarking against the EFIT code at a flat-top phase to demonstrate the method's accuracy and the effectiveness of the weight-adaptive strategy. Section 4 further evaluates the method's robustness by accurately reconstructing three full EAST discharges with different plasma shape (lower single-null, upper single-null, and double-null-like), validating its consistent performance across diverse configurations. Conclusion and discussion are covered in the last section.

2. Method

PINN integrates deep learning with physical principles to solve complex scientific and engineering problems. Unlike traditional neural networks, PINN is designed to incorporate known physical laws, typically expressed as PDEs, directly into the deep learning process [21]. In this work, the architecture of PINN is developed by embedding the governing G-S equation, the magnetic measurements and other boundary conditions into the loss function of the neural network. By minimizing this physics-informed loss, the PINN can accurately and robustly reconstruct EAST plasma equilibria by synergistically leveraging the experimental data and the fundamental physical laws, while strictly adhering to the constraints imposed by the G-S equation.

This section first introduces the EAST tokamak and its standard equilibrium reconstruction methods. Subsequently, it elaborates on the key components of our PINN approach, including the network design, the construction of loss functions, and the training strategy—with particular emphasis on the automatic weight-adaptive technique employed to enhance convergence and accuracy.

2.1. Numerical equilibrium reconstruction algorithms

As shown in Fig. 1, the EAST device is equipped with 12 PF coils for plasma generation, heating, and equilibrium control, and a pair of anti-parallel inner coils for plasma vertical instability control, leading to a total of 13 PF coil power supplies in the following calculations. Plasma equilibrium reconstruction relies on magnetic diagnostic systems, including 38 magnetics probes (blue square) for local poloidal magnetic fields measurement, 35 flux loops (red dot) for magnetic flux measurement, and Rogowski coils (not shown) directly monitor total plasma current [14,35].

Taking the widely-used EFIT code as a representative example [14], the principle of the traditional numerical algorithm for EAST plasma equilibrium reconstruction is to compute the poloidal flux function ψ in the (R,Z) plane based on the G-S equation with the toroidal current density J_ϕ constrained by the two stream functions pressure $p(\psi)$ and poloidal current $f(\psi)$. The EFIT framework, which this description follows, determines these stream functions through a least-square best fit to the available experimental data from various diagnostics including external magnetic measurements and the imposed physics constraints [8,14]:

$$\Delta^* \psi = -\mu_0 R J_\phi \quad (1)$$

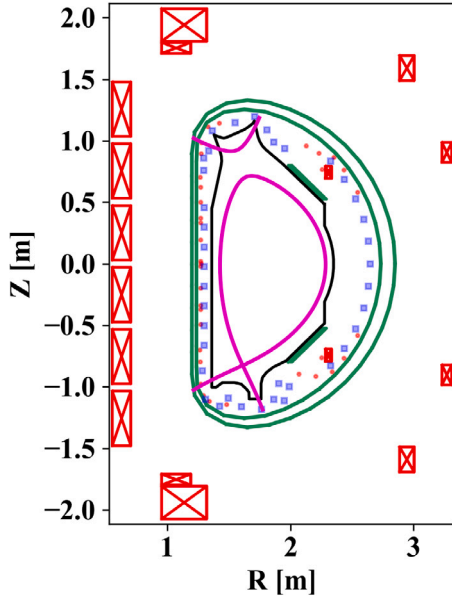


Fig. 1. EAST geometry and the location of flux loops (red dot), magnetic probes (blue square), and PF coils (red rectangular frame with cross).

$$J_\phi = Rp' + \frac{\mu_0 f f'}{4\pi^2 R} \quad (2)$$

where ψ is the poloidal flux per radian of the toroidal angle Φ enclosed by a magnetic surface, $f = \frac{2\pi R B_\phi}{\mu_0}$, B_ϕ is the toroidal magnetic field, and $\Delta^* = R^2 \nabla \cdot (\frac{\nabla}{R^2})$. The toroidal current density J_ϕ is modeled as being distributed among the elements of grid points, through the two stream functions represented in terms of a set of linear coefficients α_n and γ_n as:

$$p'(\psi) = \sum_n^{N_p} \alpha_n \psi_n^n \quad (3)$$

$$f f'(\psi) = \sum_n^{N_F} \gamma_n \psi_n^n \quad (4)$$

where the normalized flux $\psi_N = \frac{\psi - \psi_a}{\psi_b - \psi_a}$ varies between 0 and 1, ψ_0 and ψ_1 are the poloidal flux at the magnetic axis and that at the plasma boundary, respectively. The number of N_p and N_F represent the chosen order of the stream function, that can be determined depends on the amount of the available experimental data as well as on the sensitivity of the data [10]. Then, the calculated value of magnetic measurements are obtained from the Picard iterations:

$$C_j^{(m+1)}(r_j) = \sum_{n=1}^{n_{PF}} G_{c_j}^{PF}(r_j, r_{PF_n}) I_{PF_n}^{(m+1)} + \int dR' dZ' G_{c_j}^{plasma}(r_j, r') J_\phi(r', \psi^{(m)}, \alpha_n^{(m+1)}, \gamma_n^{(m+1)}) \quad (5)$$

where C_j is the calculated value of j th measurement, G_{c_j} is the appropriate Green's function corresponding to the j th measurement, $I_{PF_n}^{(m+1)}$ is the current in the n th PF coil at the $m+1$ iteration, n_{PF} is the total number of the PF coils. The first summation corresponds to the response from discrete poloidal field coils, denoted by the Green's function $G_{c_j}^{PF}$, whose second argument takes the discrete coil locations r_{PF_n} . The integral term corresponds to the response from the distributed plasma current, denoted by $G_{c_j}^{plasma}$, whose second argument runs over continuous plasma volume points r' . During the iteration cycles, α_n and γ_n are readjusted continuously, according to the available measured

data, by finding a linearized minimization

$$\chi^2 = \sum_{j=1}^{n_M} \left(\frac{M_j - C_j}{\sigma_j} \right)^2 \quad (6)$$

where M_j , C_j and σ_j denote the measured value, the calculated value and the error associated with the j th measurement, respectively. n_M is the total number of measurements. The solution is calculated using a new value of normalized flux ψ_N to calculate the new set of (α_n, γ_n) coefficients of the plasma current density model in Eq. (2). The iteration process is continued until the poloidal flux function ψ converges to a given tolerance such as $\epsilon = 10^{-3}$ for all (R,Z) grid points [35].

$$\left| \frac{\psi^{(m)} - \psi^{(m+1)}}{\psi_b - \psi_a} \right| < \epsilon \quad (7)$$

The values of (α_n, γ_n) obtained in the last iteration are adopted as the correct coefficients of the plasma current density representation. After convergence, the equilibrium parameters, such as pressure profile, current flux function, poloidal flux distribution, poloidal beta, internal inductance, the plasma shape (including the last closed flux surface, LCFS) and position can be calculated [8,35].

2.2. PINN architecture

The proposed PINN architecture is composed of three main parts as depicted in Fig. 2. The first part focuses on predicting the poloidal flux ψ , the second part derives the p' and $f f'$ by minimizing the residual of G-S equation, and the third part is to calculate the PF coil currents I_{PF} . Instead of directly getting the total poloidal flux ψ as described in Ref. [30,31], it is further divided into two distinct components: the flux generated by the plasma ψ_{plasma} and the flux produced by the PF coils ψ_{PF} . This strategy of explicitly separating the plasma and external coil contributions to the total flux is conceptually aligned with that employed in other machine-learning-based equilibrium reconstruction works, such as the method presented by McClenaghan et al. which uses basis functions as well as Green's function tables [36]. However, our approach integrates this decomposition into a PINN framework with three dedicated subnetworks and an adaptive weighting scheme, enabling end-to-end optimization without relying on precomputed basis sets.

$$\psi = \psi_{PF} + \psi_{plasma} \quad (8)$$

As illustrated in Fig. 2, ψ_{plasma} is modeled by a fully connected neural network with R and Z coordinates as inputs. The network architecture consists of 4 hidden layers, each containing 20 neurons activated by the hyperbolic tangent (tanh) function. This activation function was selected after empirical comparisons with alternatives (e.g., ReLU, sigmoid) showed superior convergence and accuracy in solving the G-S equation. And ψ_{PF} is calculated by multiplying the PF coil currents I_{PF} with the coefficient matrix G_{PF} , which is equivalent to Green function of EFIT [37].

$$\psi_{PF} = G_{I_{PF}} \times I_{PF} \quad (9)$$

Although Rogowski coils and fiber-optic current sensors [38] can be used to measure coil currents, direct incorporation of these measured values into the G-S equation leads to large residuals and convergence difficulties due to measurement uncertainties. To address this, an additional neural network is introduced to correct the coil current values. This network consists of two hidden layers with 20 neurons each, all fully connected and activated by the tanh function. It takes the measured currents from 13 sets of PF coils (including in-vessel control coils) as input and outputs their corrected values. Constrained by the physical and boundary loss terms of the PINN, this network acts as a nonlinear filter, ensuring that the output currents remain close to the measured values while better satisfying the G-S equation and various magnetic diagnostic constraints. The last part of the PINN network,

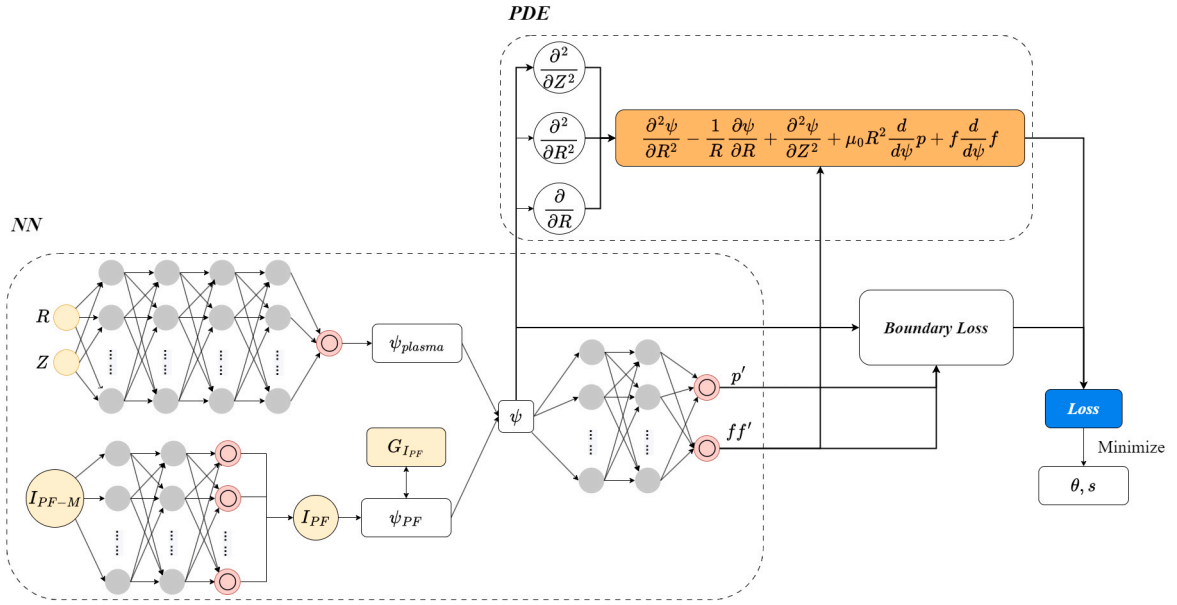


Fig. 2. The PINN architecture comprises three specialized subnetworks: magnetic flux ψ reconstruction subnetwork, plasma current profile reconstruction subnetwork, and PF coil currents reconstruction subnetwork. The network parameters θ and adaptive weights s are determined by minimizing the total loss function.

which incorporates the physical constraints of the G-S equation, takes the total poloidal flux as its input and outputs the plasma current profile functions $p'(\psi)$ and $ff'(\psi)$. This sub-network consists of two fully connected hidden layers, each containing 20 neurons with tanh activation. The dual-output architecture explicitly encodes the functional dependence of p' and ff' on ψ , as rigorously required by the G-S equation. The outputs of these subnetworks collectively define the equilibrium state. To ensure this state is both physically valid and consistent with experimental measurements, a composite loss function is constructed. Beyond standard terms for fitting diagnostics and the G-S equation, a critical innovation is the inclusion of Green's function-enhanced constraints. These constraints directly link the neural network's predicted plasma current to the magnetic signals it should produce at the diagnostic locations, thereby providing a powerful regularizer to avoid unphysical current profile solutions.

2.3. Loss functions

The loss function for the EAST plasma equilibrium reconstruction using PINN integrates two fundamental components: the physical loss, derived from the G-S equation, and the boundary loss, which incorporates magnetic diagnostic measurements. These components are further specialized into the following specific loss terms:

a. Flux loops:

$$\mathcal{L}_{FL} = \frac{1}{N_{FL}} \sum_{i_{FL}=1}^{N_{FL}} \frac{(\psi(R_{i_{FL}}, Z_{i_{FL}}) - \psi_{i_{FL}}^b)^2}{(\sigma_{i_{FL}})^2} \quad (10)$$

where N_{FL} denotes the number of flux loops, $\psi_{i_{FL}}^b$ the measured flux, $\psi(R_i, Z_i)$ the predicted flux, and σ_i the measurement uncertainty.

b. Magnetic probes:

$$\mathcal{L}_{BP} = \frac{1}{N_{BP}} \sum_{i_{BP}=1}^{N_{BP}} \frac{(B_R(R_{i_{BP}}, Z_{i_{BP}}) \cos \theta_{i_{BP}} + B_Z(R_{i_{BP}}, Z_{i_{BP}}) \sin \theta_{i_{BP}} - B_{i_{BP}}^b)^2}{(\sigma_{i_{BP}})^2} \quad (11)$$

where N_{BP} is the number of magnetic probes, $B_R(R_{i_{BP}}, Z_{i_{BP}})$ and $B_Z(R_{i_{BP}}, Z_{i_{BP}})$ the magnetic field components computed via automatic differentiation ($B_R = -\frac{1}{R} \frac{\partial \psi}{\partial Z}$ and $B_Z = \frac{1}{R} \frac{\partial \psi}{\partial R}$), $\theta_{i_{BP}}$ the probe orientation, $B_{i_{BP}}^b$ the measured field, and $\sigma_{i_{BP}}$ the uncertainty.

The uncertainties $\sigma_{i_{FL}}$ and $\sigma_{i_{BP}}$ are synthesized from both relative and absolute error components. Specifically, $\sigma_{i_{FL}}$ combines a relative error term proportional to the measured flux value with an absolute systematic error from flux loop calibration and positioning. Similarly, $\sigma_{i_{BP}}$ incorporates a relative error component scaled by the magnetic field measurement along with fixed systematic errors arising from probe orientation and calibration uncertainties. This composite error model ensures proper weighting of diagnostic measurements in the PINN optimization.

As motivated earlier, to mitigate the risk of unphysical current profiles and to strengthen the physical coupling in the inverse problem, we introduce enhanced constraints via Green's function-based response matrix analysis, systematically correlating magnetic diagnostics with p' and ff' profiles.

c. Flux loops (Green's function constrained):

$$\mathcal{L}_{G-FL} = \frac{1}{N_{FL}} \sum_{i_{FL}=1}^{N_{FL}} \frac{(\psi_{G-plasma}(R_{i_{FL}}, Z_{i_{FL}}) - \psi_{plasma}(R_{i_{FL}}, Z_{i_{FL}}))^2}{(\sigma_{i_{FL}})^2} \quad (12)$$

$$\psi_{G-plasma}(R_{i_{FL}}, Z_{i_{FL}}) = G_{in-FL} \cdot \left[Rp' + \frac{ff'}{\mu_0 R} \right] S \quad (13)$$

where $\psi_{G-plasma}(R_{i_{FL}}, Z_{i_{FL}})$ and $\psi_{plasma}(R_{i_{FL}}, Z_{i_{FL}})$ denote the plasma-contributed poloidal flux at the i th flux loop, computed via the Green's function method and the PINN method, respectively; G_{in-FL} is the Green's function matrix between points inside the LCFS and the flux loops; and S represents the area element associated with the collocation points.

d. Magnetic probes (constrained by Green's function):

$$\mathcal{L}_{G-BP} = \frac{1}{N_{BP}} \sum_{i_{BP}=1}^{N_{BP}} \frac{(B_{G-plasma}(R_{i_{BP}}, Z_{i_{BP}}) - B_{plasma}(R_{i_{BP}}, Z_{i_{BP}}))^2}{(\sigma_{i_{BP}})^2} \quad (14)$$

$$B_{G-plasma}(R_{i_{BP}}, Z_{i_{BP}}) = G_{in-BP} \cdot \left[Rp' + \frac{ff'}{\mu_0 R} \right] S \quad (15)$$

where $B_{G-plasma}(R_{i_{BP}}, Z_{i_{BP}})$ and $B_{plasma}(R_{i_{BP}}, Z_{i_{BP}})$ represent the plasma-contributed magnetic field from Green's function and PINN, respectively; and G_{in-BP} represents the Green's function matrix between points inside the LCFS and the magnetic probes.

Moving beyond the limiter as boundary approximation in Ref. [31], our algorithm explicitly computes the LCFS, providing improved

plasma boundary characterization consistent with Ref. [30]. Thus, distinct formulations of the G-S equation are employed to physically constrain randomly distributed collocated points across different plasma regions.

e. Physics Loss inside the LCFS:

$$\mathcal{L}_{p,inside} = \frac{1}{N_{pin}} \sum_{i_p=1}^{N_{pin}} \left[\left(\frac{\partial^2 \psi}{\partial R^2} - \frac{1}{R} \frac{\partial \psi}{\partial R} + \frac{\partial^2 \psi}{\partial Z^2} + \mu_0 R^2 \frac{d}{d\psi} p + f \frac{d}{d\psi} f \right)^2 \right]_{(R_{ip}, Z_{ip})_{inside}} \quad (16)$$

f. Physics loss outside the LCFS:

$$\mathcal{L}_{p,outside} = \frac{1}{N_{pout}} \sum_{i_p=1}^{N_{pout}} \left[\left(\frac{\partial^2 \psi}{\partial R^2} - \frac{1}{R} \frac{\partial \psi}{\partial R} + \frac{\partial^2 \psi}{\partial Z^2} \right) \right]_{(R_{ip}, Z_{ip})_{outside}} \quad (17)$$

During the training procedure, the pressure gradient p' and ff' evaluated at LCFS points $(R_{i_{LCFS}}, Z_{i_{LCFS}})$ should be appropriately constrained to zero or finite value, with the selection criterion determined by the plasma discharge regime. In this work, $p'(R_{i_{LCFS}}, Z_{i_{LCFS}})$ and $ff'(R_{i_{LCFS}}, Z_{i_{LCFS}})$ are constrained to zero based on chosen discharge characteristics.

g. p' loss on the LCFS:

$$\mathcal{L}_{p',LCFS} = \frac{1}{N_{LCFS}} \sum_{i_{LCFS}=1}^{N_{LCFS}} (p'(R_{i_{LCFS}}, Z_{i_{LCFS}}))^2 \quad (18)$$

h. ff' loss on the LCFS:

$$\mathcal{L}_{ff',LCFS} = \frac{1}{N_{LCFS}} \sum_{i_{LCFS}=1}^{N_{LCFS}} (ff'(R_{i_{LCFS}}, Z_{i_{LCFS}}))^2 \quad (19)$$

i. Plasma current loss:

$$\mathcal{L}_{ip} = \sum_{i_p=1}^{N_{pin}} (ip(R_{ip}, Z_{ip})_{inside} - ip_{measure})^2 \quad (20)$$

$$ip = (Rp' + \frac{ff'}{\mu_0 R})S \quad (21)$$

where $ip_{measure}$ is the measured value of the total plasma current usually by Rogowski coil.

j. Poloidal field coil current loss:

$$\mathcal{L}_{PF} = \frac{1}{N_{PF}} \sum_{i=1}^{N_{PF}} (I_{PF_i} - I_{PF_i}^b)^2 \quad (22)$$

Instead of treating the measured PF coil currents as fixed inputs to the network, we adopt an approach consistent with EFIT: refining and reconstructing them through the loss function. This methodology promotes network convergence and yields solutions that are more aligned with the G-S equation.

Since averaging the physical losses across all collocated points can lead to poor convergence by underrepresenting regions with high residuals, we introduce a focused optimization strategy. In each iteration, we compute new loss terms $\mathcal{L}_{p,insidemax}$ and $\mathcal{L}_{p,outsidemax}$ from the top 3% of the absolute values of $\mathcal{L}_{p,inside}$ and $\mathcal{L}_{p,outside}$, respectively. This approach prioritizes the most challenging points within and outside the LCFS for constraint enforcement.

The total loss function is given by:

$$\begin{aligned} \mathcal{L} = & w_{FL} \mathcal{L}_{FL} + w_{G-FL} \mathcal{L}_{G-FL} + w_{BP} \mathcal{L}_{BP} + w_{G-BP} \mathcal{L}_{G-BP} \\ & + w_{p,inside} \mathcal{L}_{p,inside} \\ & + w_{p,outside} \mathcal{L}_{p,outside} + w_{p',LCFS} \mathcal{L}_{p',LCFS} + w_{ff',LCFS} \mathcal{L}_{ff',LCFS} \quad (23) \\ & + w_{ip} \mathcal{L}_{ip} \\ & + w_{p,insidemax} \mathcal{L}_{p,insidemax} + w_{p,outsidemax} \mathcal{L}_{p,outsidemax} + w_{PF} \mathcal{L}_{PF} \end{aligned}$$

where w represents the weights for each loss component.

2.4. Weight-adaptive method

Owing to the complexity of the loss function, which comprises twelve distinct components (as in Eq. (23))—significantly more than those used in typical PINN for solving PDEs like Poisson's or Burgers' equations—and the prevalence of measurement errors in the boundary conditions, convergence becomes particularly challenging. This necessitates an adaptive method to automatically learn the optimal weights for each loss term, thereby enhancing the accuracy and robustness of the equilibrium reconstruction [39,40]. Drawing inspiration from the work of Kendall et al. [41], and noting similar applications of uncertainty quantification in fusion ML surrogates [42], this paper employs an adaptive loss balancing method. The core of this approach lies in its use of Gaussian maximum likelihood estimation to automatically update loss weights by modeling the uncertainty inherent in multi-task learning. In the context of plasma equilibrium reconstruction, the problem is formulated as a multi-objective optimization, requiring the simultaneous fitting of the G-S equation and various magnetic diagnostics. Furthermore, measurement errors introduce significant inherent uncertainties that critically impact the results. To address these challenges, we integrate the weight-adaptive method into our PINN framework for equilibrium reconstruction. The specific algorithm is detailed as follows:

First, the output u of the PINN is modeled as a Gaussian distribution, characterized by a mean given by the network's prediction $\hat{u}$ and a standard deviation represented by the uncertainty parameter σ :

$$p(u|\hat{u}(x, t; \theta)) = \mathcal{N}(\hat{u}(x, t; \theta), \sigma^2) \quad (24)$$

where θ represents the parameters of the network. Our goal is to find the parameters θ that maximize the likelihood p . Since we ultimately aim to minimize a loss function, we convert this objective by taking the negative logarithm of the likelihood:

$$\begin{aligned} -\log p(u|\hat{u}(x, t; \theta)) & \propto \frac{1}{2\sigma^2} \|u - \hat{u}(x, t; \theta)\|^2 + \log \sigma \\ & = \frac{1}{2\sigma^2} \mathcal{L}_u(\theta) + \log \sigma \end{aligned} \quad (25)$$

which decomposes into a σ -weighted loss term and a $\log \sigma$ term.

Furthermore, assuming N independent sets of outputs u_1 to u_N , each with a Gaussian distribution,

$$p(u_1, u_2, \dots, u_N | \hat{u}(x, t; \theta)) = \prod_{n=1}^N p(u_n | \hat{u}(x, t; \theta)) = \prod_{n=1}^N \mathcal{N}(\hat{u}(x, t; \theta), \sigma_n^2) \quad (26)$$

Next, we apply the negative logarithm to the likelihood,

$$\begin{aligned} -\log p(u_1, u_2, \dots, u_N | \hat{u}(x, t; \theta)) & = -\log \sum_{n=1}^N \mathcal{N}(\hat{u}(x, t; \theta), \sigma_n^2) \\ & \propto \sum_{n=1}^N \left(\frac{1}{2\sigma_n^2} \mathcal{L}_n + \log \sigma_n \right) \end{aligned} \quad (27)$$

At this point, the loss function for PINN derived through the weight-adaptive algorithm, is given by:

$$\mathcal{L}(\theta, \sigma) = \sum_{n=1}^N \frac{1}{2\sigma_n^2} \mathcal{L}_n + \log \prod_{n=1}^N \sigma_n \quad (28)$$

The objective of optimizing the loss function is to identify an optimal set of network parameters θ and loss weights σ , which subsequently serve as the parameters for the PINN surrogate model that approximates the solution to the given PDE. Here, each individual loss component is weighted by its corresponding uncertainty. Since these weights are also integrated into the total loss function, they can be adaptively optimized and dynamically adjusted. For instance, if the uncertainty σ_x for a specific loss term $\mathcal{L}_x$ decreases, its associated weight increases, thereby amplifying its contribution to the total loss. Concurrently, the inclusion of the $\log \sigma_x$ term acts as a counterbalance, preventing this particular loss term from dominating the optimization

process and ensuring that other, more difficult-to-converge terms are not neglected. In practice, we parameterize the network to predict the log-variance, $s := \log \sigma^2$:

$$\mathcal{L}(\theta, s) = \sum_{n=1}^N \left(\frac{e^{-s_n}}{2} \mathcal{L}_n + s_n \right) \quad (29)$$

This parameterization offers greater numerical stability than directly regressing the variance σ^2 , as it prevents the risk of division by zero in the loss function.

Substituting the terms from Eq. (23) into Eq. (29) yields the loss function augmented with the weight-adaptive algorithm:

$$\begin{aligned} \mathcal{L} = & \frac{e^{-s_{FL}}}{2} \mathcal{L}_{FL} + \frac{e^{-s_{G-FL}}}{2} \mathcal{L}_{G-FL} + \frac{e^{-s_{BP}}}{2} \mathcal{L}_{BP} + \frac{e^{-s_{G-BP}}}{2} \mathcal{L}_{G-BP} \\ & + \frac{e^{-s_{p,inside}}}{2} \mathcal{L}_{p,inside} \\ & + \frac{e^{-s_{p,outside}}}{2} \mathcal{L}_{p,outside} + \frac{e^{-s_{p',LCFS}}}{2} \mathcal{L}_{p',LCFS} + \frac{e^{-s_{ff',LCFS}}}{2} \mathcal{L}_{ff',LCFS} \\ & + \frac{e^{-s_{ip}}}{2} \mathcal{L}_{ip} \\ & + \frac{e^{-s_{p,insidemax}}}{2} \mathcal{L}_{p,insidemax} + \frac{e^{-s_{p,outsidemax}}}{2} \mathcal{L}_{p,outsidemax} + \frac{e^{-s_{PF}}}{2} \mathcal{L}_{PF} \\ & + \frac{s_{FL}}{2} + \frac{s_{G-FL}}{2} + \frac{s_{BP}}{2} + \frac{s_{G-BP}}{2} + \frac{s_{p,inside}}{2} + \frac{s_{p,outside}}{2} + \frac{s_{p',LCFS}}{2} \\ & + \frac{s_{ff',LCFS}}{2} + \frac{s_{ip}}{2} + \frac{s_{p,insidemax}}{2} + \frac{s_{p,outsidemax}}{2} + \frac{s_{PF}}{2} \\ & \propto e^{-s_{FL}} \mathcal{L}_{FL} + e^{-s_{G-FL}} \mathcal{L}_{G-FL} + e^{-s_{BP}} \mathcal{L}_{BP} \\ & + e^{-s_{G-BP}} \mathcal{L}_{G-BP} + e^{-s_{p,inside}} \mathcal{L}_{p,inside} \\ & + e^{-s_{p,outside}} \mathcal{L}_{p,outside} + e^{-s_{p',LCFS}} \mathcal{L}_{p',LCFS} + e^{-s_{ff',LCFS}} \mathcal{L}_{ff',LCFS} \\ & + e^{-s_{ip}} \mathcal{L}_{ip} \\ & + e^{-s_{p,insidemax}} \mathcal{L}_{p,insidemax} + e^{-s_{p,outsidemax}} \mathcal{L}_{p,outsidemax} + e^{-s_{PF}} \mathcal{L}_{PF} + s_{FL} \\ & + s_{G-FL} + s_{BP} + s_{G-BP} + s_{p,inside} + s_{p,outside} + s_{p',LCFS} + s_{ff',LCFS} \\ & + s_{ip} \\ & + s_{p,insidemax} + s_{p,outsidemax} + s_{PF} \end{aligned} \quad (30)$$

2.5. Training process

Departing from conventional fixed-grid schemes, the proposed method trains the ψ_{plasma} network using 10,000 spatially random $P(R, Z)$ collocation points generated via Latin Hypercube Sampling [43], thereby mitigating inherent discretization and truncation errors [44]. The training incorporates not only these points but also experimental magnetic measurements and the complete set of Green's function matrices. These matrices physically couple the collocation points, magnetic diagnostics (flux loops and magnetic probes), and PF coils. The inclusion of the full matrix set is paramount, as omitting any component relaxes the physical constraints on the solution and introduces significant stochasticity into the reconstruction.

Complementing the data strategy, the choice of optimization algorithm is equally crucial for efficiently training the deep neural network, which in this work consists of 3276 trainable parameters initialized using the Xavier scheme [45]. Among the prevalent optimizers, Adaptive Moment Estimation (Adam) [46] and Limited-memory Broyden-Fletcher-Goldfarb-Shanno (L-BFGS) [47] represent two prominent yet distinct approaches, each with unique advantages [48,49]. To address the ill-conditioning issues inherent in training PINN for plasma equilibrium reconstruction, this work employs a hybrid strategy that leverages the complementary strengths of both Adam and L-BFGS [50].

Building upon this hybrid optimizer strategy, the training follows a precisely ordered, cyclic procedure. It starts with a 10,000-iteration warm-up using the Adam optimizer, during which the LCFS is initially set to the limiter wall. The core cycle then begins: first, the L-BFGS optimizer refines the network parameters for 1000 iterations. A systematic

grid search [51] across cycle lengths from 100 to 2000 confirmed that 1000 iterations achieve an optimal trade-off between cost and accuracy. Next, the Adam optimizer is employed for 50 iterations, specifically to adaptively adjust the weights of the loss terms, which are treated as trainable parameters. Following this weight optimization, the LCFS is updated based on the network's latest flux (ψ) predictions, and the collocation points are dynamically reclassified. This cycle repeats, with the learning rate for Adam being reduced based on the maximum physical residuals $L_{p,insidemax}$ and $L_{p,outsidemax}$: initially set to 5×10^{-3} , it is lowered to 2×10^{-3} once these residuals fall below 1×10^{-2} , and further to 1×10^{-3} when below 1×10^{-3} . The entire process terminates once both key physical residuals drop below 1×10^{-4} or after 500 LCFS update cycles, indicating successful convergence. The complete algorithm is outlined in Algorithm 1.

Algorithm 1 weight-adaptive PINN for equilibrium reconstruction

Input: $P(R, Z)$, $P_{FL}(R, Z)$, $P_{BP}(R, Z)$, $P_{limiter}(R, Z)$, ψ_{FL}^b , B_{BP}^b , σ_{FL} , σ_{BP} , G_{P-FL} , G_{P-BP} , G_{P-PF} , $i_{p,measure}$, I_{PF}^b
Output: θ , s

- 1: Initialize θ_0 and s_0 ;
- 2: Initialize P_{in} , P_{out} , P_{LCFS} by $P(R, Z)$ and $P_{limiter}(R, Z)$;
- 3: Initialize $lr_{ADAM} = 5 \times 10^{-4}$, $lr_{ADAM_0} = 1 \times 10^{-3}$
- 4: **for** each $i \in [1, N_{adam}]$ **do**
- 5: $\theta_{i+1} \leftarrow \text{Adam}_0(\mathcal{L}(\theta_i, s_0))$;
- 6: **end for**
- 7: **for** each $j \in [1, iter_{L-BFGS}]$ **do**
- 8: $\theta_{j+1} \leftarrow \text{L-BFGS}(\mathcal{L}(\theta_j, s_j))$;
- 9: $s_{j+1} \leftarrow \text{Adam}(\mathcal{L}(\theta_{j+1}, s_j))$;
- 10: Update P_{in} , P_{out} , $P_{LCFS} \leftarrow \psi_{NN(\theta)}(P(R, Z))$
- 11: **if** $\mathcal{L}_{p,insidemax} < 1 \times 10^{-2}$ and $\mathcal{L}_{p,outsidemax} < 1 \times 10^{-2}$ **then**
- 12: $lr_{ADAM} = 2 \times 10^{-4}$
- 13: **else if** $\mathcal{L}_{p,insidemax} < 1 \times 10^{-3}$ and $\mathcal{L}_{p,outsidemax} < 1 \times 10^{-3}$ **then**
- 14: $lr_{ADAM} = 1 \times 10^{-4}$
- 15: **else if** $\mathcal{L}_{p,insidemax} < 1 \times 10^{-4}$ and $\mathcal{L}_{p,outsidemax} < 1 \times 10^{-4}$ **then**
- 16: Break
- 17: **end if**
- 18: **if** $j = 500$ **then**
- 19: Break
- 20: **end if**
- 21: **end for**
- 22: **return** θ , s

3. Validation

To validate the effectiveness of the proposed weight-adaptive PINN method, we benchmark its reconstruction results against those from the EFIT code. It is important to note that while EFIT serves as a widely adopted and extensively validated numerical tool in tokamak research, providing a reliable baseline for comparison, it does not yield a perfectly precise reference solution due to its inherent dependence on predefined parameterizations for the plasma current profiles. This limitation aligns with the recognized challenge that the plasma current profile cannot be accurately reconstructed using only external magnetic diagnostics [6]. The comparison presented herein is therefore intended to demonstrate the consistency and capability of our PINN approach against an established conventional method, rather than to assert absolute accuracy. Nevertheless, EFIT's solutions, particularly concerning the LCFS and the X-point location derived from the reconstructed flux surfaces, are generally considered robust and trustworthy, which justifies its use as a baseline and the subsequent comparison of X-point coordinates in Section 4.

Based on this rationale, we conducted equilibrium reconstruction on a recent EAST discharge (#157868), selecting a representative time slice at 8.0 s during the flat-top phase as a benchmark. At this time point, we systematically compared the performance of the vanilla PINN

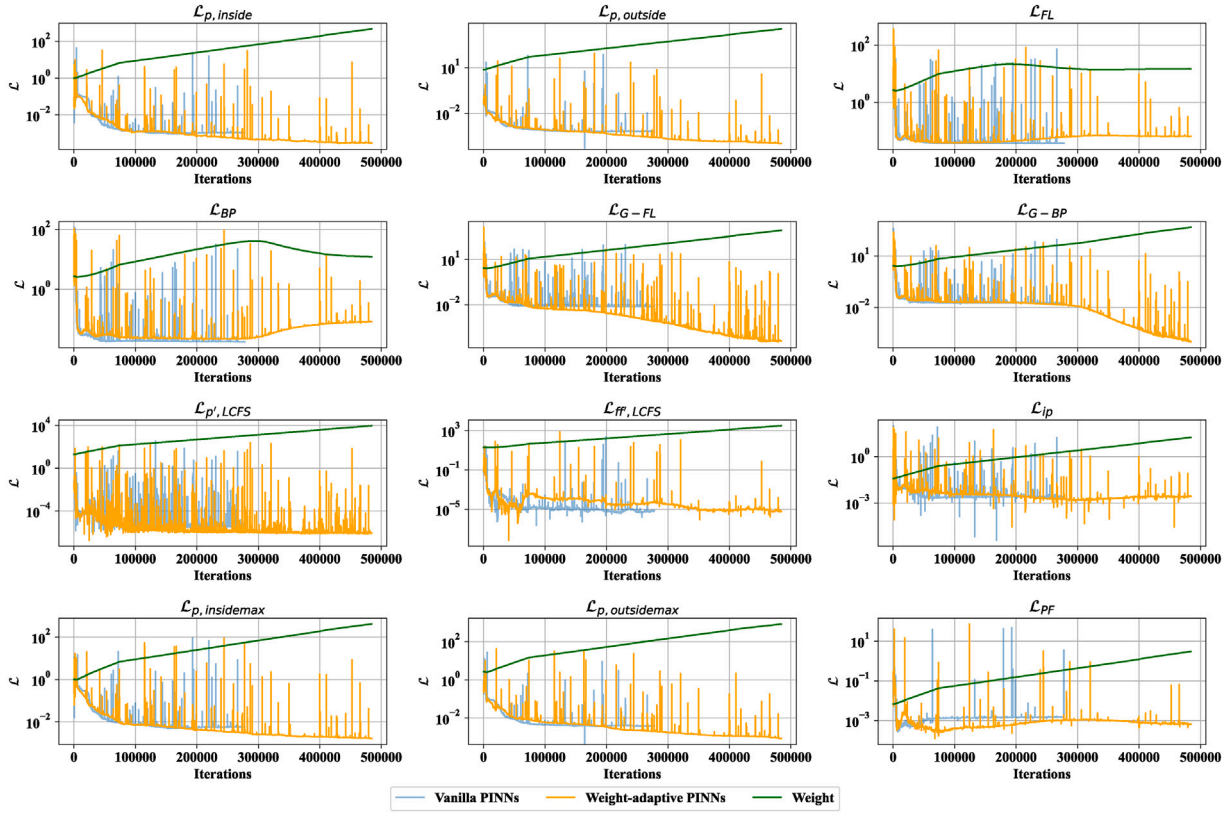


Fig. 3. EAST discharge #157868 at $t = 8$ s: Convergence of individual loss functions of vanilla PINN(blue), weight-adaptive PINN(orange) and their adaptive weights(green) are shown.

against our weight-adaptive variant. All experiments were performed under identical conditions to ensure a fair comparison, with key hyperparameters – including the number of Adam iterations, Adam learning rate, maximum L-BFGS iterations, and the initial learning rate for adaptive weight parameters – held constant across runs. The models were implemented using the PyTorch framework.

As shown in Fig. 3, we present the individual loss components and the evolution of the adaptive weights for both models. During the initial training phase, all loss weights exhibited an increasing trend, coinciding with a rapid decline in the loss values. Around iteration 7500, the weight-adaptive module triggered a learning rate reduction from 5×10^{-4} to 2×10^{-4} , which consequently slowed the rate of increase in the loss weights. At approximately iteration 200,000, the vanilla PINN encountered premature convergence, halting training early at iteration 277,955 due to an irreconcilable conflict between the measurement errors from magnetic diagnostics and the physical constraints. In contrast, the weight-adaptive PINN responded by reducing the weights for the conflicting terms ($\mathcal{L}_{FL}$ and $\mathcal{L}_{BP}$), which facilitated continued optimization until the predefined 500 LCFS cycles were completed. A comparison of the final loss values reveals that the weight-adaptive PINN achieved lower values than the vanilla PINN across all terms except for $\mathcal{L}_{FL}$ and $\mathcal{L}_{BP}$. This demonstrates that the method's strategy of adaptively reducing the weights of the error-containing measurement items effectively suppressed overfitting to the boundary conditions and concurrently promoted the convergence of the remaining terms.

Figs. 4 and 5 presents a comprehensive comparison of the plasma equilibrium reconstruction results at $t = 8$ s between the weight-adaptive PINN and the vanilla PINN, both benchmarked against the EFIT code. The left column compares the global reconstruction error relative to EFIT. The weight-adaptive PINN achieves an error below 3×10^{-3} , whereas the vanilla PINN shows significantly larger discrepancies, indicating a failure to converge to a physically consistent solution. The middle column displays the reconstructed magnetic flux surfaces. Both

PINN methods produce flux surfaces that appear visually similar to EFIT's results, with the weight-adaptive PINN showing slightly better alignment. To quantify the accuracy at critical shape control positions, we evaluated the intersection points C1–C6 between the six control segments and the LCFS. These points are defined at poloidal directions of $0, 0.25\pi, 0.75\pi, \pi, 1.25\pi$, and 1.75π radians from the magnetic axis. Here, C1 corresponds to 0 rad (outboard midplane to the right). The labeling proceeds sequentially in the counterclockwise direction: C2 at 0.25π (upper-right), C3 at 0.75π (upper-left), C4 at π (inboard left), C5 at 1.25π (lower-left), and C6 at 1.75π (lower-right). The X-point coordinates were also evaluated alongside these points. Quantitative analysis at C1–C6 and X-point (Table 1) shows that although the vanilla PINN yields a slightly smaller maximum error (0.776 cm), its overall boundary deviation and X-point error are notably larger. In comparison, the weight-adaptive PINN achieves a maximum error of 0.915 cm and reduces the X-point deviation to only 0.17 cm, thereby meeting the required control-accuracy specification. The right column illustrates the distribution of physics-informed loss residuals ($\mathcal{L}_{p,inside}$ and $\mathcal{L}_{p,outside}$). For the weight-adaptive PINN, the maximum residual remains below 0.06, with nearly 90% of points below 0.02, confirming effective convergence of the Grad-Shafranov equation. In contrast, the vanilla PINN exhibits high and unstructured physics residuals that stagnated during training (Fig. 3), explaining its inferior core reconstruction.

Fig. 6 provides a detailed comparison of the reconstructed PF coil currents (top row) and internal plasma current profile functions (bottom row). The top row shows that both PINN methods reconstruct PF coil currents (I_{PF}) that agree reasonably well with EFIT results and experimental measurements, with the weight-adaptive PINN achieving a marginally better fit. The bottom row, however, reveals stark differences in the current profile functions p' and $f f'$. The weight-adaptive PINN produces smooth, physically reasonable profiles that approach zero at the LCFS, showing strong consistency with EFIT. Conversely, the vanilla PINN yields p' and $f f'$ profiles which exhibit

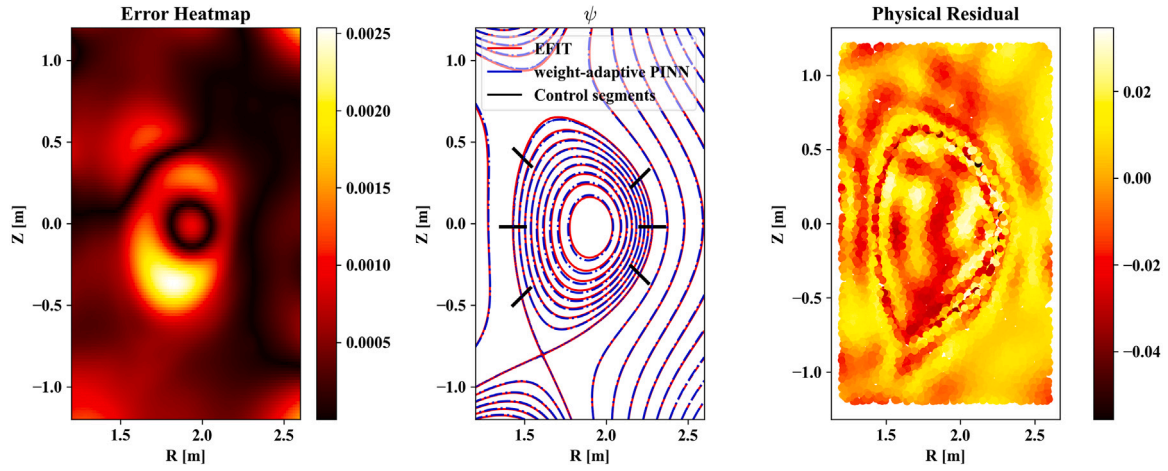


Fig. 4. EAST discharge #157868 at $t = 8$ s: the reconstruction error between weight-adaptive PINN and EFIT(left), a comparison of the magnetic flux surfaces with the LCFS(middle), and the distribution of the physics-informed loss residuals from the weight-adaptive PINN(right).

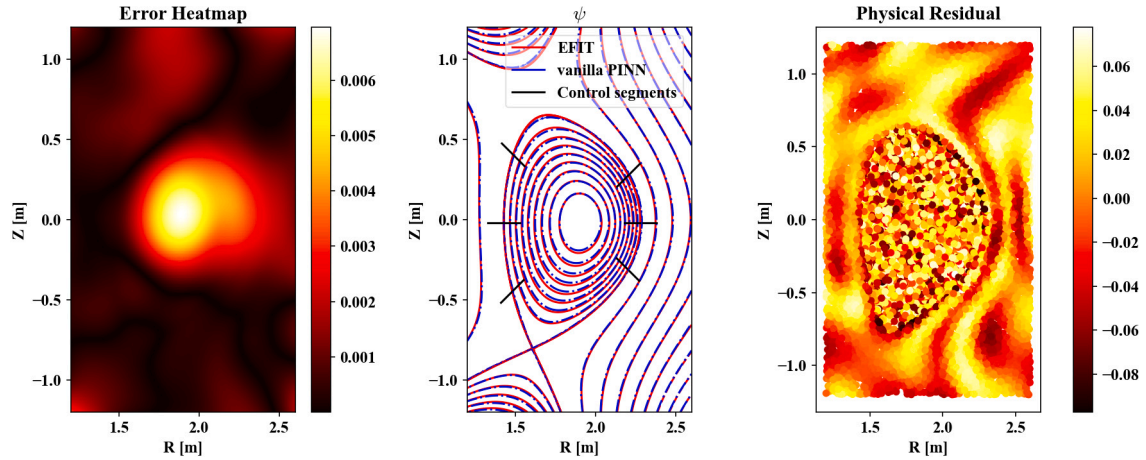


Fig. 5. EAST discharge #157868 at $t = 8$ s: the reconstruction error between vanilla PINN and EFIT(left), a comparison of the magnetic flux surfaces with the LCFS(middle), and the distribution of the physics-informed loss residuals from the vanilla PINN(right).

Table 1

Boundary error obtained from weight-adaptive PINN and vanilla PINN [cm].

	C1	C2	C3	C4	C5	C6	X-point
Weight-adaptive PINN	0.118	0.029	0.915	0.165	0.032	0.086	0.170
Vanilla PINN	0.776	0.755	0.630	0.001	0.481	0.371	0.679

large deviations from the EFIT results and do not converge to the expected behavior, particularly in the core region of the plasma. This comparison highlights the critical role of the proposed weight-adaptive strategy. The vanilla PINN suffers from premature convergence due to unresolved conflicts between measurement noise and physical constraints, resulting in high physics residuals and inaccurate core profiles. Our weight-adaptive PINN mitigates this by dynamically balancing loss terms, suppressing overfitting to noisy diagnostics, and prioritizing convergence of the physics equation, ultimately achieving accurate and physically consistent reconstruction, particularly in the plasma core.

For a further quantitative analysis, Table 2 presents a benchmark comparison of the equilibrium reconstruction results. The Root Mean Square Error (RMSE), Relative Mean Absolute Error (RMAE), and the coefficient of determination (R^2) are defined as follows:

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (31)$$

Table 2

Quantitative comparison of weight-adaptive PINN and vanilla PINN with measurements and EFIT.

	RMSE		RMAE		R^2	
	w-a	Vanilla	w-a	Vanilla	w-a	Vanilla
Flux loops	9.23×10^{-4}	6.69×10^{-4}	0.63%	0.46%	99.9754%	99.9871%
Magnetic probes	1.02×10^{-3}	6.07×10^{-4}	0.75%	0.44%	99.9863%	99.9951%
ψ	6.52×10^{-4}	1.77×10^{-3}	0.58%	1.32%	99.9397%	99.8165%
I_{PF}	3.24×10^4	3.85×10^4	69.6%	56.0%	99.6953%	99.5710%

$$RMAE = \frac{1}{n} \sum_{i=1}^n \frac{|y_i - \hat{y}_i|}{|y_i|} \quad (32)$$

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (33)$$

The elevated RMSE and RMAE for the PF coil current I_{PF} warrant a detailed explanation. This phenomenon is typical for EAST and arises from the specific operational characteristics of its coil set. PF13 is a fast vertical displacement stabilization control coil, whose current is typically two orders of magnitude smaller (on the order of 10^1 to 10^2 A) than those of the primary shaping coils (PF1-PF12, on the order of 10^4 to 10^6 A). Consequently, even a modest absolute error in reconstructing PF13's current (e.g., 10^2 A) results in a very high relative error because

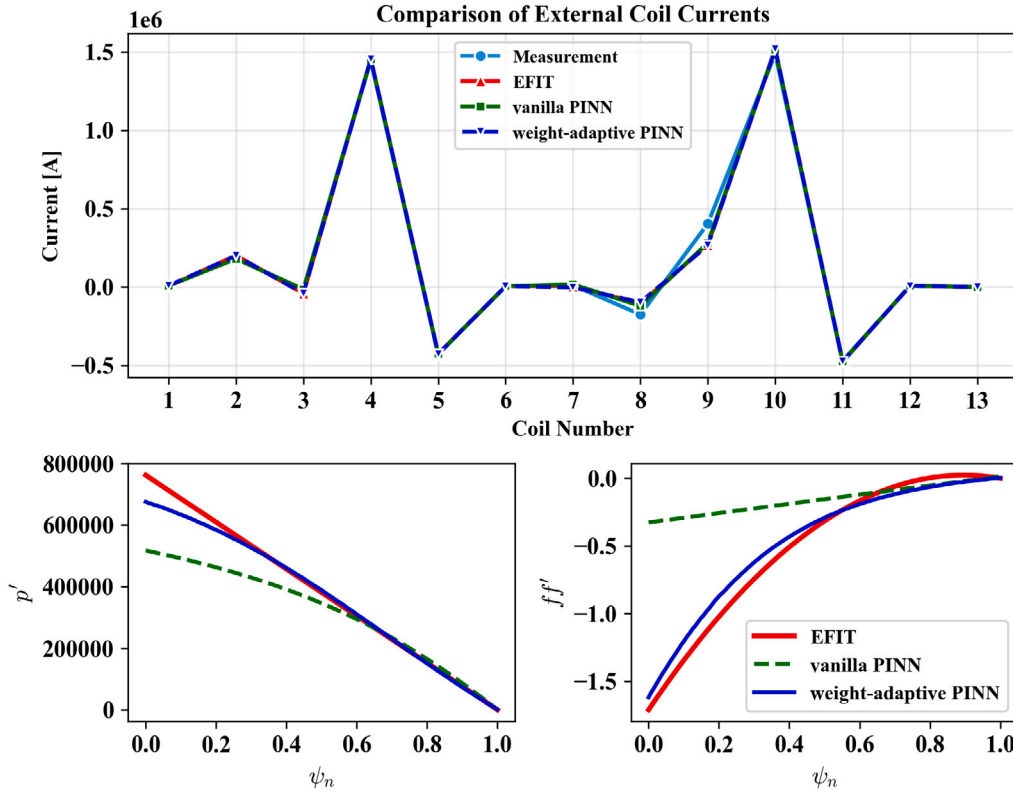


Fig. 6. EAST discharge #157868 at $t = 8$ s: Comparison of PF coil currents(top) and internal plasma current profiles(bottom) reconstructed by the weight-adaptive PINN, vanilla PINN and EFIT.

the denominator (the measured value) is of similar magnitude. This single high relative error disproportionately inflates the overall RMAE. Conversely, the substantial RMSE is primarily driven by larger absolute errors (on the order of 10^6 A) from a few major coils with the highest currents. It is crucial to note that this statistical metric reflects the scale of corrections applied to reconcile measurements with the G-S equation, not a failure in reconstruction, as the reconstructed I_{PF} values show excellent agreement with both EFIT and measurements (Fig. 6). Furthermore, while the vanilla PINN yields lower RMSE and RMAE and a higher R^2 for flux loops and magnetic probes, this indicates overfitting to these potentially noisy diagnostic signals. Consequently, its reconstructed magnetic flux ψ shows a larger deviation from the EFIT reference. In contrast, the weight-adaptive PINN achieves a more accurate and physically consistent ψ reconstruction that aligns closely with EFIT, demonstrating the effectiveness of its adaptive strategy in mitigating overfitting and enhancing solution fidelity.

Overall, these results collectively demonstrate that the weight-adaptive PINN successfully overcomes the fundamental limitations of the vanilla PINN and achieves an accurate and reliable plasma equilibrium reconstruction by fitting the magnetic measurement boundary conditions and the G-S equation.

4. Experimental results

To systematically evaluate the robustness and accuracy of the weight-adaptive PINN method across diverse plasma configurations, it was applied to reconstruct equilibria throughout three complete EAST discharges, each representing a distinct magnetic topology: shot #157868 (lower single-null), shot #115287 (upper single-null), and shot #122264 (double-null-like). For these full-discharge comparisons with the established EFIT code, we utilized the lower-sampling-rate equilibrium data from the PCS-EAST database. While sparse, this data's temporal resolution is fully sufficient to capture the complete equilibrium evolution across all discharge phases, which is the focus of our

methodology assessment. The method's performance was then assessed through comprehensive comparisons with EFIT over the full discharge durations of these three configurations.

4.1. Lower single-null discharge (#157868)

Fig. 7 presents a quantitative analysis of the reconstruction reliability over the complete discharge duration of shot #157868, tracking the evolution of plasma current I_p and key shape parameters – including the major radius R_0 , minor radius a , and elongation κ – followed by X-point coordinates (R_{xpt} , Z_{xpt}). The trajectories reconstructed by both methods demonstrate remarkable consistency from start to end. Specifically, the plasma current reconstructed by the weight-adaptive PINN shows excellent agreement with EFIT, with discrepancies remaining below 3 kA across all operational phases. More importantly, the reconstructed shape parameters exhibit high fidelity throughout the discharge, with maximum differences of 0.5 cm for the major radius, 1 cm for the minor radius, and 0.057 for the elongation. The X-point positions also maintain close alignment, with a maximum positional error of 0.7 cm. This sustained sub-centimeter accuracy in determining both the global shape parameters and the critical X-point location underscores the spatial precision and temporal reliability of the weight-adaptive PINN framework for full-discharge analysis.

To further validate the method's effectiveness across distinct operational regimes, eight time slices were randomly selected from the ramp-up, flat-top, and ramp-down phases ($t = 0.9$ s, 2 s, 4 s, 5 s, 6 s, 7.5 s, 8.5 s, and 10.7 s). Fig. 8 compares the poloidal flux ψ distributions reconstructed by EFIT and the weight-adaptive PINN at these eight time slices. The results show high consistency in the magnetic flux surfaces, including the shape of the LCFS and the internal magnetic structures, across all tested scenarios. The consistent agreement demonstrates the method's capability to accurately capture plasma equilibria under different discharge conditions, confirming its robust performance across diverse phases of tokamak discharge.

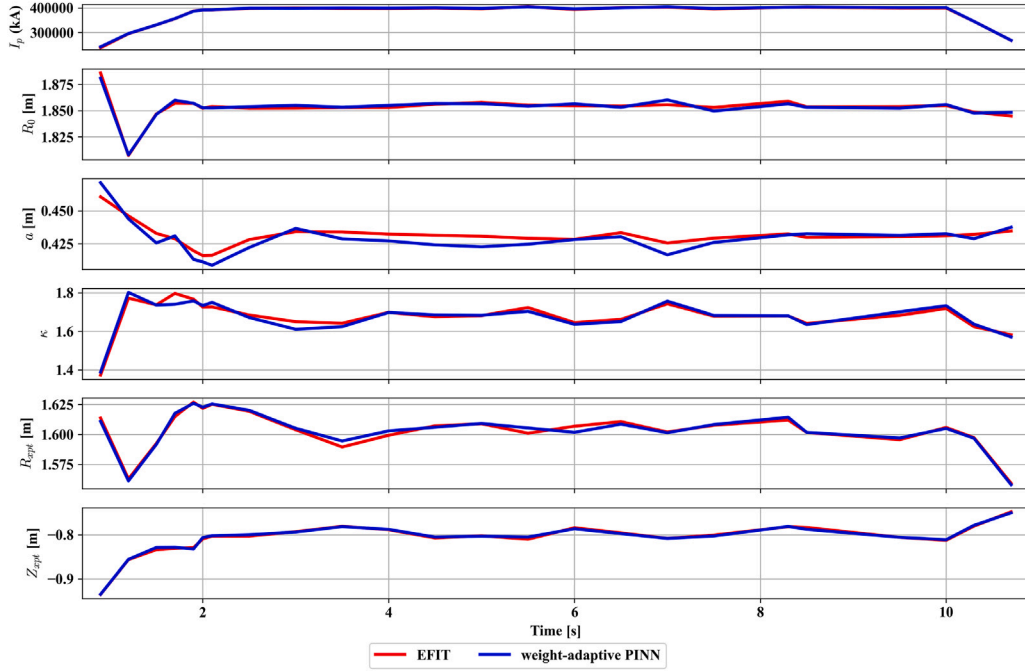


Fig. 7. EAST discharge #157868: Comparison of plasma current (I_p), major radius (R_0), minor radius (a), elongation (κ) and X-point coordinates (R_{xpt} , Z_{xpt}) between EFIT and the weight-adaptive PINN reconstruction across 25 representative time slices spanning the ramp-up, flat-top, and ramp-down phases.

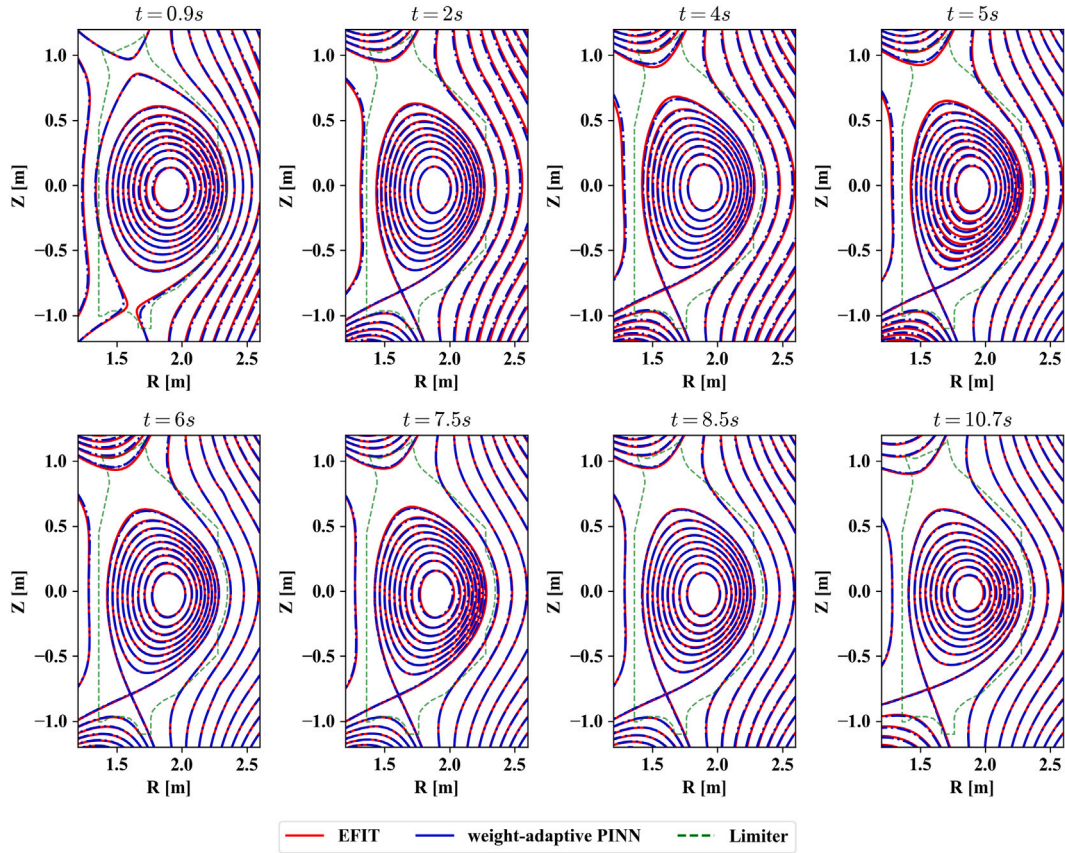


Fig. 8. EAST discharge #157868 : Comparison of the poloidal flux ψ from EFIT and the weight-adaptive PINN at 8 randomly selected time slices.

4.2. Upper single-null discharge (#115287)

For the upper single-null discharge #115287, Fig. 9 illustrates the evolution of plasma current and shape parameters throughout the

entire discharge. The weight-adaptive PINN reconstruction maintains excellent agreement with EFIT, with plasma current discrepancies consistently below 3.2 kA. Shape parameters show similar fidelity to the lower single-null case, with maximum deviations of 0.36 cm for major

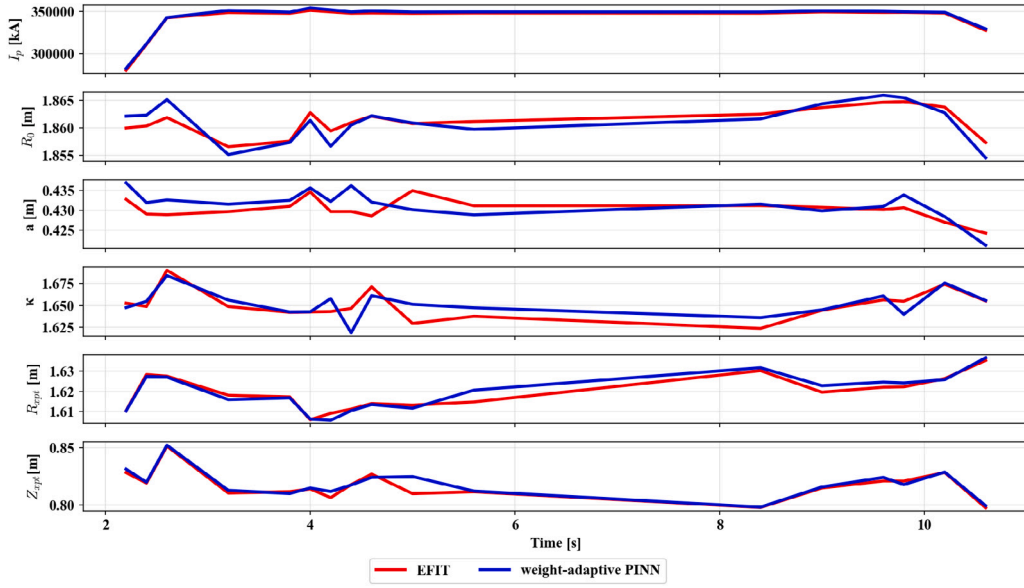


Fig. 9. EAST discharge #115287: Comparison of plasma current (I_p), major radius (R_0), minor radius (a), elongation (κ) and X-point coordinates (R_{xpt} , Z_{xpt}) between EFIT and the weight-adaptive PINN reconstruction across 18 representative time slices spanning the ramp-up, flat-top, and ramp-down phases.

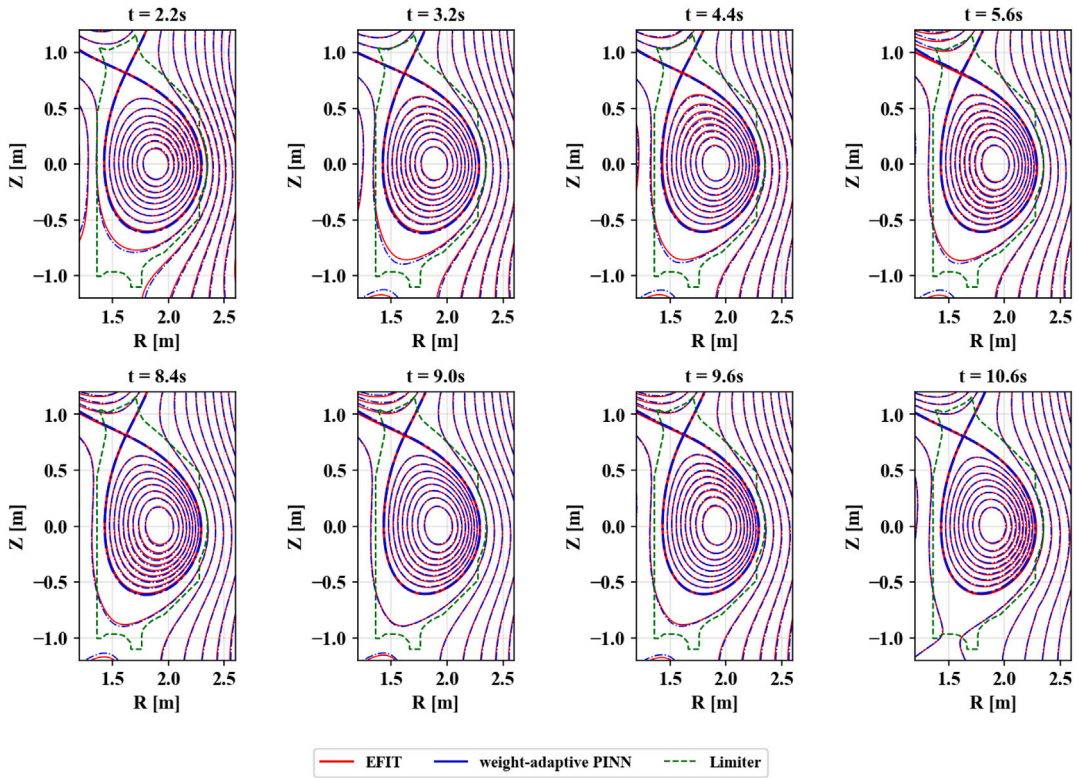


Fig. 10. EAST discharge #115287 : Comparison of the poloidal flux ψ from EFIT and the weight-adaptive PINN at 8 randomly selected time slices.

radius, 0.7 cm for minor radius, and 0.028 for elongation. The X-point coordinates are accurately reconstructed with a maximum error of 0.64 cm, demonstrating the method's capability to handle the inverted magnetic geometry of upper single-null configurations.

Eight representative time slices spanning the discharge phases ($t = 2.2$ s, 3.2 s, 4.4 s, 5.6 s, 8.4 s, 9.0 s, 9.6 s, and 10.6 s), were selected for detailed flux surface analysis. As shown in Fig. 10, the reconstructed poloidal flux distributions exhibit excellent agreement with EFIT across all time points. The method successfully captures the flux near the upper X-point and accurately reproduces the LCFS

shape throughout the discharge evolution. These results validate the framework's adaptability to different magnetic field orientations while maintaining consistent accuracy.

4.3. Double-null-like discharge (#122264)

The double-null-like discharge #122264, with its inherently more complex magnetic topology featuring two active X-points, presents a stringent test for the reconstruction method. Fig. 11 shows the

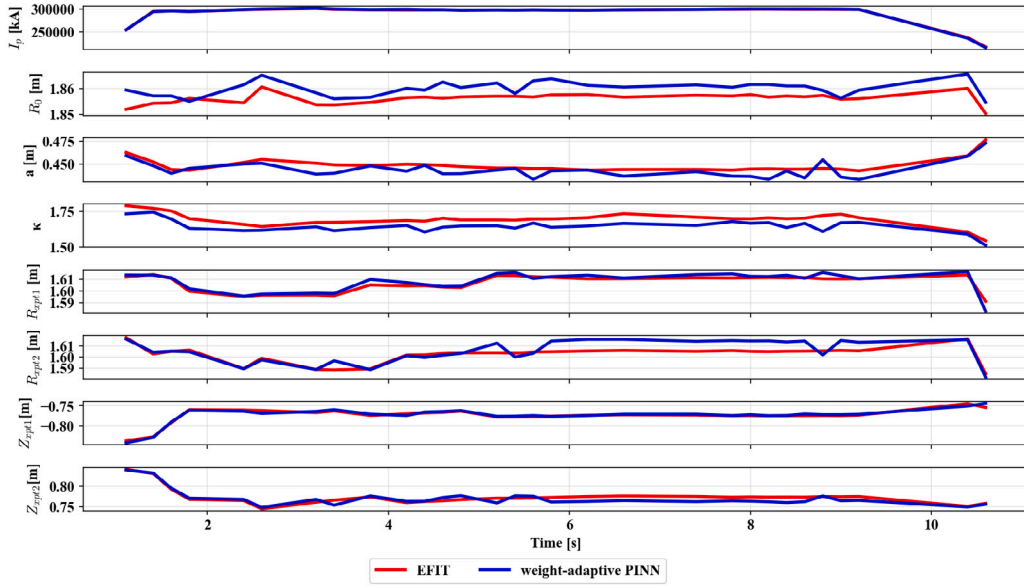


Fig. 11. EAST discharge #122264: Comparison of plasma current (I_p), major radius (R_0), minor radius (a), elongation (κ) and X-point coordinates (R_{xpt} , Z_{xpt}) between EFIT and the weight-adaptive PINN reconstruction across 17 representative time slices spanning the ramp-up, flat-top, and ramp-down phases.

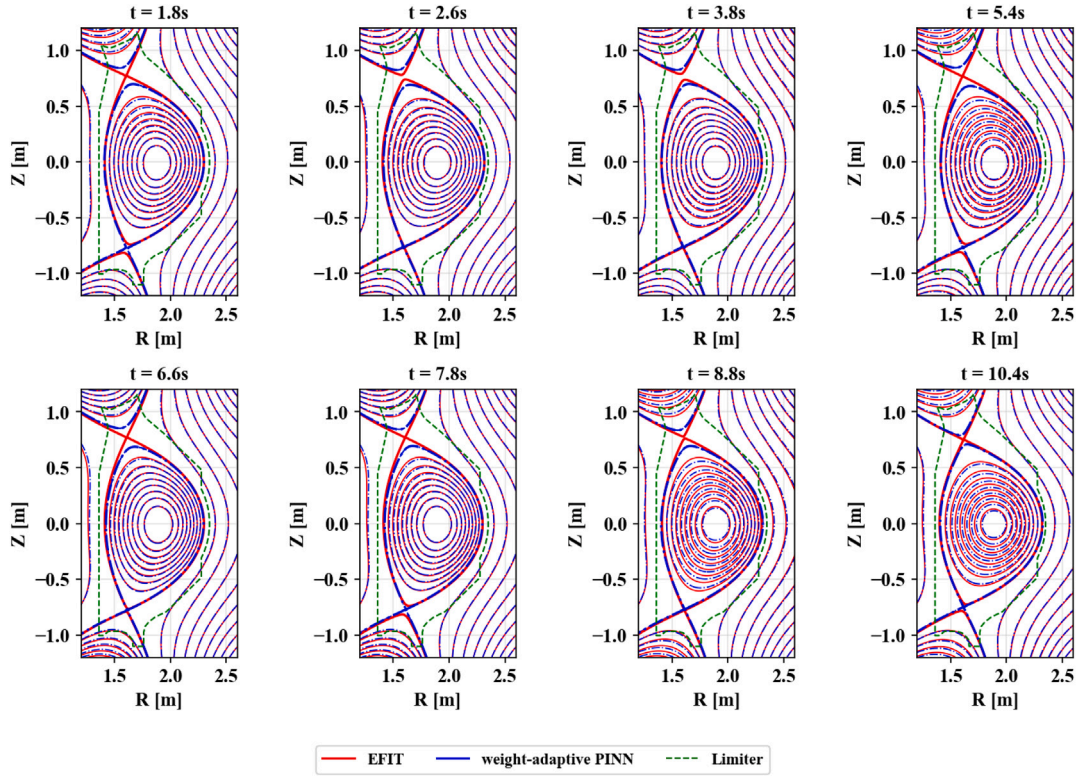


Fig. 12. EAST discharge #122264 : Comparison of the poloidal flux ψ from EFIT and the weight-adaptive PINN at 8 randomly selected time slices.

temporal evolution of key parameters compared with EFIT. The weight-adaptive PINN maintains good agreement on the global plasma current, with discrepancies below 3.5 kA. However, reconstruction accuracy for shape parameters is reduced compared to the single-null cases, with maximum errors of 0.7 cm for the major radius, 1.2 cm for the minor radius, and 0.12 for the elongation. A notable performance asymmetry is observed in the reconstruction of the two X-points: while the lower X-point is localized with a maximum error of 0.8 cm, consistent with single-null performance, the upper X-point shows a significantly larger

maximum deviation of 1.6 cm. This indicates that the current framework, while robust for dominant magnetic structures, exhibits varying precision when resolving multiple, simultaneous magnetic nulls.

Eight time slices ($t = 1.8$ s, 2.6 s, 3.8 s, 5.4 s, 6.6 s, 7.8 s, 8.8 s, and 10.4 s) were analyzed to examine flux surface reconstruction across different discharge phases. As demonstrated in Fig. 12, the method successfully captures the global double-null topology and the evolution of the magnetic structure. However, a closer inspection reveals a systematic tendency: the reconstruction shows higher fidelity in the region

of the lower X-point, with the shape of the lower flux surfaces and the lower divertor leg closely matching EFIT. In contrast, the poloidal flux contours near the upper X-point exhibit greater discrepancy, particularly in the sharpness of the null and the shape of the upper separatrix. This suggests that the present implementation of the physics-informed constraints may naturally favor the reconstruction of one dominant null region in a complex magnetic field, a point for potential refinement in future work aimed at perfectly symmetric double-null scenarios.

These results demonstrate that the weight-adaptive PINN framework achieves good agreement with EFIT for equilibrium reconstruction across the entire discharge cycle. The method performs consistently well for both upper and lower single-null configurations, accurately recovering key global parameters and flux surface details through ramp-up, flat-top, and ramp-down phases. While the reconstruction for the double-null-like configuration maintains acceptable accuracy in global metrics, it reveals a current limitation in equally resolving fine magnetic structures at both X-points, with the upper null showing larger positional errors. This contrast highlights the method's robustness in handling standard single-null topologies and pinpoints a specific challenge in perfectly symmetric, complex multi-null scenarios. Overall, the framework demonstrates strong potential as a versatile tool for equilibrium reconstruction, with its performance on prevalent single-null configurations being particularly robust, while also providing clear direction for future refinements aimed at more intricate magnetic geometries.

5. Conclusion and discussion

In this work, a novel weight-adaptive PINN framework has been successfully developed and applied to plasma equilibrium reconstruction on the EAST tokamak. The proposed architecture comprises three dedicated subnetworks that collectively predict the poloidal flux, the current profile functions (p' and $f f'$), and PF coil currents, providing an integrated approach to the equilibrium reconstruction task. As demonstrated in Sections 3 and 4, this approach accurately reconstructs key equilibrium features – including the plasma shape, X-point coordinates, and internal current profiles – across diverse discharge phases, showing remarkable consistency with the established EFIT code and experimental measurements throughout a full discharge.

The proposed framework offers three key advantages over conventional numerical methods and vanilla PINN approaches. Firstly, it eliminates the reliance on predefined current profile assumptions by adaptively resolving the p' and $f f'$ functions through a dedicated subnetwork, providing enhanced capability for handling complex current distributions. Secondly, it refines the treatment of PF coils by integrating a dedicated subnetwork that corrects the measured coil currents, thereby reconciling experimental measurements with the physical constraints of the G-S equation and yielding more self-consistent solutions. Thirdly, the incorporation of an adaptive loss weighting scheme, combined with a hybrid Adam-LBFGS optimizer, automatically balances the multiple constraints from physics and diagnostics. This strategy effectively mitigates conflicts arising from measurement uncertainties, prevents premature convergence, and contributes significantly to achieving robust training and improved reconstruction accuracy. Collectively, these features – combined with the mesh-free nature of the framework – facilitate efficient equilibrium analysis and enhance reconstruction capabilities.

Despite these promising results, several challenges remain to be addressed to advance the practical implementation of this method. A primary limitation is the accurate reconstruction of equilibria with non-monotonic current profiles, which is essential for characterizing advanced plasma regimes. Enhancing diagnostic integration through the real-time assimilation of internal measurements, such as Motional Stark Effect (MSE) data, could substantially improve the boundary condition constraints for core profile reconstruction. Furthermore, error propagation analysis indicates that diagnostic uncertainties in magnetic

measurements can induce variations in the reconstructed results, highlighting the need for a Bayesian uncertainty quantification framework to provide reliable error estimates.

The framework developed in this work is, in principle, not restricted to the EAST tokamak. Its core architecture – integrating the Grad-Shafranov equation with external magnetic diagnostics through a trainable, adaptive PINN – constitutes a general methodology for solving free-boundary equilibrium reconstruction problems. This is supported by related work applying similar PINN concepts to other devices such as JET [31]. Successful adaptation to a new tokamak would require incorporating device-specific parameters: primarily the Green's function matrices for its unique set of magnetic diagnostics and PF coils, along with their precise geometrical layouts and calibration data. With these inputs, our three-subnetwork architecture and training pipeline could be directly applied. The current manuscript focuses on establishing and thoroughly validating the method for EAST, thereby providing a foundational case study and a reusable template for extensions to future devices.

In conclusion, the weight-adaptive PINN framework provides a reliable, effective, and promising alternative for analyzing EAST discharges. By integrating physical laws with measured diagnostic data in a flexible and self-consistent manner, this approach offers improved accuracy and robustness for plasma equilibrium reconstruction. With ongoing developments to incorporate diverse diagnostics and adapt to complex plasma conditions, the framework holds significant potential to support equilibrium analysis for future fusion devices such as ITER and China Fusion Engineering Test Reactor (CFETR).

CRediT authorship contribution statement

Wenbin Liao: Writing – original draft, Validation, Software, Methodology, Investigation, Formal analysis, Data curation. **Zhengping Luo:** Writing – review & editing, Methodology, Investigation, Formal analysis, Data curation. **Yao Huang:** Methodology, Investigation, Funding acquisition. **Yuehang Wang:** Project administration, Methodology, Investigation, Funding acquisition. **Zijie Liu:** Methodology, Investigation, Funding acquisition. **Kai Wu:** Methodology, Investigation, Funding acquisition. **Dalong Chen:** Methodology, Investigation. **Shengze Cai:** Methodology, Investigation. **Bingjia Xiao:** Supervision, Methodology, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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